

RETAIL SALES ANALYTICS

SweetFoot.Co Sales Performance Case Study

PivotTable & PivotChart Driven Analysis of Employee, Product and Transaction Performance
— July 2022

Prepared by: Prince Maheta

Project Repository:
github.com/princemaheta2627-glitch

Category: Data Analytics / Data Science Portfolio Project — Case Study 3

Tool Used: Microsoft Excel (PivotTables, PivotCharts, Slicers, Calculated Fields, Ranking Formulas)

1. Executive Summary

This project analyzes one month (July 2022) of point-of-sale transaction data for SweetFoot.Co, a hypothetical footwear retailer, to identify the best-performing employee, the best-performing product, and the average transaction value — the three KPIs defined at the outset of the project. The analysis was built entirely in Excel using PivotTables, PivotCharts, slicers and ranking formulas, moving from a single raw transaction log to a set of decision-ready summary views.

The workflow mirrors a standard business analytics pipeline: define KPIs, ingest raw transactional data, clean/structure it, aggregate it along multiple dimensions (employee, product, day of week), rank the segments, and visualize the results for stakeholders. This report documents the objective, dataset, methodology and key findings, and is included in my portfolio to demonstrate applied data analysis and business-insight generation using Excel.

2. Business Objective & KPIs

Objective: Sales for a month were tracked to determine the best performing employee, best performing product, and the average transaction value.

KPI	Description	Formula
Transaction Value Average	Average value spent by customers per transaction	Total Revenue / No. of Transactions
Product Contribution	Identify high vs. low performing products and their revenue share	Product Revenue / Total Revenue
Employee Sales	Find the most productive salesperson and their contribution	Salesperson Revenue / Total Revenue

3. Dataset Description

Source: SweetFoot.Co sales log for July 2022 — 66 recorded transactions (79 line items, since some transactions include multiple products) capturing Date, Day of week, Transaction number, Salesperson, Item, Quantity and Sales Amount.

Attribute	Detail
Period	1 – 30 July 2022
Transactions	66
Line items	79
Salespeople	Daksh, Naitik, Vishwas
Products	Shoes, Slippers, Sandals
Total Revenue	\$12,650
Total Units Sold	140

4. Tools & Techniques Used

Category	Techniques Applied
Data Structuring	Raw transaction log converted into an Excel Table for clean, filterable input
Aggregation	PivotTables for Sum, Average, Max and Min of Sales Amount by employee, product and day
Ranking & Indexing	RANK() and percentage-of-average formulas to compare employee performance
Percentage Contribution	Calculated "% of Grand Total" fields for product and day-of-week contribution
Cross-tabulation	Multi-level PivotTables (employee × day, product × day)
Interactivity	Slicers connected to PivotTables/PivotCharts for on-the-fly filtering
Visualization	PivotCharts (clustered column) for employee, product and day-wise comparisons

5. Analysis & Findings

5.1 KPI #1 — Average Transaction Value

Average Transaction Value = Total Revenue ÷ No. of Transactions = \$12,650 ÷ 66 = \$191.67 per transaction.

Breaking this down by salesperson (Sales_Summary pivot) reveals meaningful differences in average ticket size, not just total revenue:

	Data	
salesperson	Sum of Sales amount	Average of Sales amount2
Daksh	\$ 4,010.00	\$ 148.52
Naitik	\$ 4,590.00	\$ 158.28
Vishwas	\$ 4,050.00	\$ 184.09
Total Result	\$ 12,650.00	\$ 162.18

Figure 1: Total and average sales amount per salesperson (Sales_Summary pivot).

- Vishwas has the highest average transaction value at \$184.09 — 13.5% above the overall average — despite not having the highest total revenue.
- Naitik's average transaction value (\$158.28) sits close to the overall average (97.6% of it).
- Daksh has the lowest average transaction value (\$148.52, 91.6% of average), suggesting more frequent but lower-value sales.

	Data		
salesperson	Sum of Sales amount	Min of Sales amount2	Sum of Sales amount
Daksh	\$ 4,010.00	\$ 50.00	\$ 4,010.00
Naitik	\$ 4,590.00	\$ 40.00	\$ 4,590.00
Vishwas	\$ 4,050.00	\$ 45.00	\$ 4,050.00
Total Result	\$ 12,650.00	\$ 40.00	\$ 12,650.00

Figure 2: Maximum and minimum single-transaction values per salesperson (Sales_Statistics pivot).

The spread between each salesperson's largest and smallest sale is wide (e.g., Vishwas ranges from \$45 to \$600), indicating high day-to-day variability in individual transaction size across all three salespeople.

5.2 KPI #2 — Product Contribution

Item-level revenue and quantity analysis shows a clear product hierarchy:

Product	Revenue	% of Total	Units Sold
Shoes	\$7,130	56.4%	57
Slippers	\$3,430	27.1%	50
Sandals	\$2,090	16.5%	33

Row Labels	Sum of No. of items
Sandals	33
Daksh	8
Naitik	14
Vishwas	11
Shoes	57
Daksh	12
Naitik	22
Vishwas	23
Slippers	50
Daksh	25
Naitik	19
Vishwas	6
Total Result	140

Figure 3: Units sold by product, broken down by salesperson (Item_Quantity_Analysis pivot).

- Shoes is the clear flagship product — over half (56.4%) of all revenue from just 57 of the 140 units sold (41% of unit volume), implying a materially higher average price point (~\$125/unit vs. ~\$63–69/unit for the other two lines).
- Slippers sell in similar volume to Shoes (50 vs. 57 units) but generate less than half the revenue, confirming it is a lower-price, higher-frequency item.
- Sandals is the smallest contributor on both revenue and volume, making it the weakest of the three product lines this month.

5.3 KPI #3 — Employee Sales Performance & Ranking

	Data		RANK
salesperson	TOTAL SALES	PERCENTAGE	
Daksh	\$ 4,010.00	91.58%	3
Naitik	\$ 4,590.00	97.59%	1
Vishwas	\$ 4,050.00	113.51%	2
Total Result	\$ 12,650.00	100.00%	

	Data		RANK
Item	Sum of Sales amount	Sum of Sales amount2	
1 Sandals	\$ 2,090.00	16.52%	
2 Shoes	\$ 7,130.00	56.36%	
Slippers	\$ 3,430.00	27.11%	
Total Result	\$ 12,650.00	100.00%	

	Data	
Day	Sum of Sales amount	Sum of Sales amount2
Monday	\$ 1,905.00	15.06%
Tuesday	\$ 2,115.00	16.72%
Wednesday	\$ 1,545.00	12.21%
Thursday	\$ 1,610.00	12.73%
Friday	\$ 2,620.00	20.71%
Saturday	\$ 2,855.00	22.57%
Total Result	\$ 12,650.00	100.00%

Figure 4: Employee ranking by total sales, plus product and day-of-week contribution breakdown (Employee_Ranking sheet).

Rank	Employee	Total Sales	% of Total Revenue
1	Naitik	\$4,590	36.3%
2	Vishwas	\$4,050	32.0%
3	Daksh	\$4,010	31.7%

- Naitik is the top-performing salesperson by total revenue, contributing 36.3% of the month's sales.
- Revenue is fairly evenly distributed across the three-person team (a ~4.6 point spread between highest and lowest), suggesting consistent overall staffing performance rather than one outlier employee.
- Ranking by total revenue and ranking by average transaction value tell different stories: Vishwas ranks 2nd on total revenue but 1st on average ticket size, while Naitik ranks 1st on total revenue but is only at-par on ticket size — indicating Naitik's lead comes from transaction frequency rather than up-selling.

5.4 Day-of-Week & Category Trends

Row Labels	Sum of Sales amount
Daksh	\$ 4,010.00
Sandals	\$ 610.00
Shoes	\$ 1,500.00
Slippers	\$ 1,900.00
Naitik	\$ 4,590.00
Sandals	\$ 805.00
Shoes	\$ 2,615.00
Slippers	\$ 1,170.00
Vishwas	\$ 4,050.00
Sandals	\$ 675.00
Shoes	\$ 3,015.00
Slippers	\$ 360.00
Total Result	\$ 12,650.00

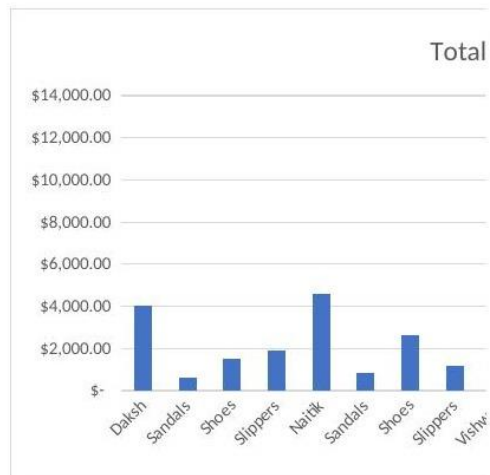
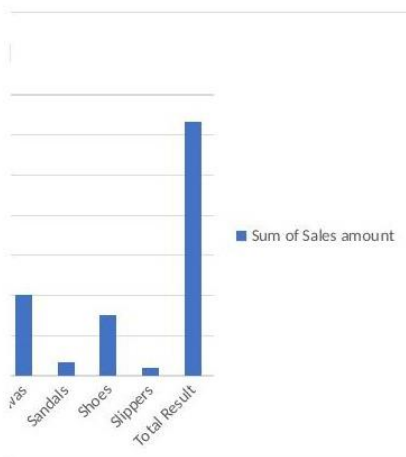


Figure 5: Total sales overview — revenue broken down by salesperson and product line (PivotChart).



This shape represents a slicer. Slicers are supported in Excel 2010 or later.

If the shape was modified in an earlier version of Excel, or if the workbook was saved in Excel 2003 or

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Figure 6: Salesperson revenue by day of week (PivotChart).

Day	Revenue	% of Total
Saturday	\$2,855	22.6%
Friday	\$2,620	20.7%
Tuesday	\$2,115	16.7%
Monday	\$1,905	15.1%
Thursday	\$1,610	12.7%
Wednesday	\$1,545	12.2%

- Weekend trading (Friday + Saturday) accounts for 43.3% of the month's total revenue despite being only 2 of 6 trading days — a strong case for prioritizing staffing and stock levels on those days.
- Wednesday is consistently the slowest day (12.2% of revenue), which could flag an opportunity for a mid-week promotion.

6. Skills Demonstrated (Relevant to Data Science Roles)

- KPI definition — translating a business question into measurable, formula-backed metrics before touching the data.

- Data structuring — converting a flat transaction log into an analysis-ready table.
- Group-by aggregation — PivotTables performing the same role as pandas `groupby().agg()` across multiple dimensions (employee, product, day).
- Statistical summarization — sum, average, max, min used together to understand both central tendency and variability.
- Ranking & comparative analysis — `RANK()` and percentage-of-average indexing to fairly compare performance across employees.
- Percentage/contribution analysis — calculated fields expressing each segment's share of the whole, the Excel equivalent of `normalize=True` in a pandas `value_counts()/groupby` call.
- Dashboard interactivity — PivotCharts linked to slicers, mirroring filter controls in BI tools like Power BI or Tableau.
- Insight communication — translating pivoted numbers into concrete, decision-relevant business recommendations (staffing, product focus, promotions).

7. Conclusion

This case study demonstrates a complete KPI-driven analytics workflow: defining business questions, structuring raw transactional data, aggregating it across employee/product/day dimensions with PivotTables, ranking performance, and visualizing results with PivotCharts and slicers. The three original KPIs were fully answered — an average transaction value of \$191.67, Shoes as the dominant product (56.4% of revenue), and Naitik as the top-performing salesperson (36.3% of revenue) — along with additional findings on weekday seasonality and price-point differences across product lines. This project is included in my portfolio to demonstrate practical, business-ready data analysis capability using Excel as the analytical engine.

8. Project Repository

The complete Excel workbook for this case study is available on GitHub:

<https://github.com/princemaheta2627-glitch>